

ARKO SAHA

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CAREER SUMMARY / RESEARCH ALIGNMENT

Data Scientist with an MSc in Data Science and a Mechatronics Engineering background, specializing in machine learning, predictive analytics, scalable ETL pipelines, and data-driven system design. Experienced across healthcare, industrial automation, integrated solutions, sports analytics, and business intelligence domains, with hands-on expertise in EHR systems, MLOps, computer vision, and data warehousing. Passionate about applied AI, foundation models, temporal embeddings, intelligent data systems, and translating complex data into impactful real-world solutions.

RESEARCH INTEREST

Applied Machine Learning (ML), Data Engineering, Deep Learning (DL), Time-Series Analytics, Intelligent Data Systems, Big Data Architectures, Computer Vision, Predictive analysis, Explainable/Responsible AI, MLOps, Optimization & classification, Large Language Models (LLMs), Reinforcement Learning (RL), Regression analysis, Retrieval Augmented Generation (RAG), Foundation Models, Electronic Health Records (EHR) Analytics, OMOP Common Data Model, Time-series & Temporal Dynamics Modelling, Learned Numerical Embeddings, Point-process Models, Time-aware Transformers

EDUCATION

MSc in Data Science

Sep 2022 – Nov 2023

University of Roehampton, London, UK

- **Scholarship:** International excellence scholarship awarded by the University of Roehampton.
- **Dissertation: Beyond instinct: data-driven cricket projection.** Incorporated mathematical models, machine learning, and interactive dashboards with KPIs to enhance player evaluation, performance forecasting, and in-game decision-making. Designed and implemented a data warehousing solution to consolidate cricket statistics from ESPN Cricinfo and Cricsheet.org, building ETL pipelines that processed 400k+ ball-by-ball records and created structured fact tables organized by player, position, and match dimensions to support predictive analytics.
- **Coursework:**
 - *Creating data visualizations with COVID-19 data using python and dashboarding using Tableau.* Leveraged Python for data visualization with COVID-19 data and implemented Tableau for creating dynamic dashboards in a project focused on presenting comprehensive insights into the pandemic's impact.
 - *A Deep Learning Ensemble Approach to Brain Tumor Early Detection from MRI Images.* Employed a deep learning ensemble approach for the early detection of brain tumors from MRI images in this project, utilizing technologies such as TensorFlow and Python.

BSc in Mechatronics Engineering

Mar 2014 – Dec 2018

Rajshahi University of Engineering & Technology, Bangladesh

- **Scholarship:** Academic excellence scholarship awarded by the Education Board Bangladesh.
- **Thesis:** An Artificial Neural Network Based Approach to the Estimation of Transmission Line Fault Location.

RELEVANT PROFESSIONAL EXPERIENCE

Freelance Data Analyst – Independent Contractor | Fiverr

Jan 2024 – Present

- Designed and deployed predictive ML models using Python (scikit-learn, pandas) to forecast trends in supply chain management, resulting in a 20% improvement in inventory accuracy for a mid-sized automation company.
- Architected automated ETL pipelines integrating multi-source data into centralized warehouse, enabling unified reporting and reducing data retrieval time by about 60% for client analytics projects.
- Delivered production-ready analytics dashboards using SQL, Power BI, and DAX to visualize key performance indicators (KPIs) for a small-sized startup powering data-driven executive decision making and a 15% increase in operational efficiency.

- Built dimensional data models (star and snowflake schemas) for retail clients, optimizing query performance and supporting complex analytical requirements across sales, inventory, and customer data domains.

Graduate Support Analyst Intern | University of Roehampton

Sep 2023 – Nov 2023

- Performed large-scale EDA and data validation pipeline utilizing Python, pivot tables, MS Excel, MS Power BI, and DAX to enhance graduate data accuracy and efficiency, yielding a 30% increase in accuracy and efficiency.
- Optimized graduate data processing pipeline to process over 7000 student records daily through meticulous data preprocessing and analysis, reducing processing time by at least 15% and ensuring data integrity for improved student outcomes.
- Collaborated with cross-functional teams to optimize student record management systems, streamlining workflows and providing timely reports to support academic and placement initiatives.

Medical Documentation Specialist | Augmedix Limited

Aug 2020 – Aug 2022

- Assisted doctors by documenting patient interactions in the electronic health record (EHR) during care to streamline documentation processes and improve efficiency. Mastered navigation of multiple EHR systems (EPIC, CERNER, eClinicalWorks) across Pediatrics, Gastroenterology, and Primary Care.
- Reviewed medical records for accuracy and completeness, collaborating with providers to update patient histories/diagnoses and conducting regular audits to ensure regulatory compliance.

Automation Support Engineer | Sincos Automation Technologies Limited

Jan 2019 – Mar 2020

- Designed and deployed computer vision inspection systems using image processing algorithms, reducing manufacturing defects by 25% and increasing throughput by 14% by eliminating bottlenecks while reducing production costs by approximately 10%.
- Automated daily operations and workflows with secured access utilizing MS Power BI, VBA, and DAX expertise and delivered actionable KPIs with a secure live dashboard – achieving a 34% increase in efficiency.

TECHNICAL PROFICIENCY

- **Data Engineering & ETL:** Data warehousing, Data Modeling, ETL/ELT Pipelines, Apache Airflow (DAGs), Dimensional Modeling (Star/Snowflake Schema), Data Integration, Snowflake, SQL Server, MySQL, PySpark.
- **Feature Engineering & Big Data:** SQL (Complex Queries, CTEs, Window Functions, Stored Procedures, Optimization), Relational databases, NoSQL, MongoDB, SQLite, Google Cloud Platform (GCP).
- **Model Development & Analytics:** Python (NumPy, Scikit-learn, Pandas, Matplotlib, Plotly, Seaborn), TensorFlow, PyTorch, Statistical Analysis, Exploratory Data Analysis, A/B testing.
- **MLOps & Infrastructure:** Linux, Docker, Git (Version Control), Flask, HTML, CSS, Bootstrap, ML System Design & Operations, Agile Methodology.
- **BI, Reporting & Visualization:** Power BI, Tableau, Advanced Excel (Formulae, DAX, XLOOKUP, Pivot Table, Power Query, VBA), Looker Studio, Figma, Strategic Planning, Market Research, Presentation.
- **Project Management & CRM:** Customer Relationship Management (CRM), Microsoft Dynamics 365.

SELECTED PROJECTS

- **FICO-Quant: Production-Scale Credit Risk Bucketing & Optimization Toolkit.**
Engineered a modular FICO bucketing framework that optimizes segmentation strategies (Equal Width, Equal Frequency, IV-based), achieving high discriminatory power (0.85 Gini) and 100% monotonicity in default rate progression.
- **Containerized Weather Data Ingestion Pipeline.**
Engineered a production-grade ingestion layer using Apache Airflow and Astro to orchestrate real-time weather ETL from Open-Meteo into PostgreSQL via PostgresHook and dBeaver.
- **End-to-End MLOps: Predictive Analytics & Asset Valuation Engine.**
Architected an asset-pricing engine for natural gas contracts, automating P&L calculations and trading strategy optimization via ARIMA-based time-series pipeline and an interactive Streamlit dashboard.

PUBLICATIONS

Jafor, A. U., Puja, P. D., Biswas, S., Saha, A., Mithu, K., & AbdullahAlMamun, K. (2026). Early Prediction of Postprandial Glycemic Response in Gestational Diabetes Using Continuous Glucose Monitoring and

Gradient Boosting Models. *Data Modeling*, 7(1), 1-7.

Haque, J., Munna, A. H., Ayon, A. S. R., Shaitee, Z. R., & Saha, A. (2020). A comprehensive overview of the pertinence and possibilities of bioactive glass in the modern biological world. *J. Biomater*, 4, 23.

LEADERSHIP EXPERIENCE

Student Representative, School of Arts & Digital Industries, University of Roehampton Oct 2022 – Sep 2023

Student Ambassador, Recruitment & Student Outreach Team, University of Roehampton Nov 2022 – Sep 2023

RELEVANT CERTIFICATIONS (SELECTED)

- **Quantitative Research** (JPMorgan Chase / Forage, 2024) | **Power BI Simulation** (PwC / Forage, 2023)
- **Data Warehouse Concepts** (Udemy, 2025) | **Snowflake: Manipulation & Intro** (Datacamp, 2025)
- **SQL: Intermediate & Intro** (Datacamp, 2025) | **Data Analytics Consulting** (KPMG / Forage, 2023)
- **Predictive Analytics using ML** (edX, 2021) | **Microsoft Power BI Masterclass** (Coursera, 2020)

LANGUAGES & REFERENCES

Languages: English (Proficient), Bengali (Native). **References:** Available upon request.